Yixiao Wang

CV

Email:yixiao.wang@duke.edu Tel: +1(919)201-8680 Homepage: Yixiao Wang Homepage

EDUCATION

Duke University, Graduate School Aug. 2024 - May. 2026 (Expected)
M.S., Statistical Science GPA: 4.0/4.0, Ranking: 1/45

University of Science and Technology of China, School of the Gifted Young Sep. 2020 - Jul. 2024
B.S., Mathematics and Applied Mathematics **(Outstanding Graduate)**
Major in Probability and Statistics track GPA (WES Converted): 3.91/4.0, Ranking: 9/92

RESEARCH INTERESTS

Machine Learning Theory, **Functional Data Analysis**, and Their Applications.

RELEVANT COURSEWORK

Mathematics & Statistics: Probability Theory (Original & Advanced, Honors, both A+), Advanced Probability Theory (A+), Mathematical Statistics (Original & Advanced, Honors, both A+), Bayesian Analysis, Time Series Analysis (A), Stochastic Processes (A-), Predictive Inference (A), Statistical Inference (A), Functional Analysis (A-), Real Analysis (A-), Complex Analysis, Mathematical Analysis I (A+), II (A), III (A-), Abstract Algebra (A-), Linear Algebra I (A+), II (A-), Fundamentals of Algebra (A), Differential Equations (A+), Operations Research, Differential Geometry (A), Fundamentals of Geometry (A).

Computer Science: Machine Learning Theory & Algorithms (A), Data Structures and Databases (A), C Programming (A+), R Programming (A), Python Programming (A), MATLAB (A-), Linux (A-).

Physics: Mechanics (A-), Optics (A), Electromagnetism (A), Atomic Physics (A+), Thermodynamics (A+), Basic Circuit Theory (A), University Physics Laboratory I (A-).

RESEARCH

Trimmed Mean for Partially Observed Functional Data | Bachelor's Thesis Feb. 2024-May. 2024
Supervised by Associate Professor Xiaohong Lan from USTC.

- Defined the Trimmed Mean for partially observed functional data.
- Proved strong consistency of depth function for partially observed functional data and the trimmed mean.
- Conducted numerical simulations experiment and corresponding R code.
- Thesis: arXiv | Code: GitHub.

PROJECTS

Airbnb Price Prediction in New York City| Course Project for *Machine Learning* Nov. 2024

- Predicted Airbnb prices using property information and achieved top 5 out of 137 participants.
- Developed an advanced stacking solution, outperforming all original submissions.
- Report: PDF | Code: GitHub.

Basketball Breakdown: NCAA Shiny App| Course Project for *R Language* Dec. 2024

- Designed and developed the Competition Analysis and Schedule pages.
- Integrated all pages into a cohesive shiny app.
- Enhanced user experience with animations for loading, smoother page navigation, and partial optimizations for the Prediction page.
- App: NCAA Data Archive | Report: PDF | Code: GitHub.

LSTM-GRU Hybrid Network | Course Project for *Time Series Analysis* Jun. 2024

- Cleaned and preprocessed the PM2.5 pollution data from Chengdu.
- Proposed an LSTM-GRU hybrid network architecture, inspired by GoogLeNet, for PM2.5 time series forecasting, which reduced the error rate by 30% compared to LSTM.
- Report: PDF | Code: GitHub.

TEACHING

Linear Algebra B1 | Teaching Assistant Mar. 2023- Jul. 2023

- Built the course homepage and served as the group leader, managing a course group with over 100 students.
- Organized weekly problem-solving sessions during weekends, spent over 40 hours of personal teaching time, and engaged with more than 200 participants, showcasing strong teaching and communication skills.
- Independently completed solution sets for over 100 post-course exercises and provided analysis and answers to all past exam papers.
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PRESENTATION

Probability Theory Seminar | Presenter April. 2023

- Delivered a special presentation with more than 10 fellow students, focusing on random matrix methods.
- The slides are available here.

INTERNSHIP

Thermo Fisher Scientific | Data Cleaning Internship Sep. 2023- Dec. 2023

- Cleaned and categorized more than 2000 customer data queries and matched them with Excel.
- Wrote Excel functions to add wildcards to the dataset to be cleaned, significantly improving team efficiency.
- Summarized the data cleansing techniques and conducted team training sessions.

AWARDS & HONORS

Outstanding Graduate from the University of Science and Technology of China (USTC), Class of 2024 **(Top 20%).**
China Petroleum Scholarship (2023): Awarded to **only 3** students in the School of the Gifted Young at USTC.
Excellent Teaching Assistant (Spring 2023): Ranked in the **top 3** at USTC, among over 1000 TAs.
Silver Prize for Outstanding Student Scholarship (2022 and 2021): Awarded to the **top 15%** of students at USTC.
First Prize in the Chinese Mathematical Competitions for University Students (2022 and 2021): Ranked in the **top 1%** of all participants.
Excellent President of the School Club (2022).
QiangWeiFengGongDeYu (Diligence and Moral Conduct) Scholarship(2022).
Bronze Prize for Outstanding Freshmen Scholarship (2020): Awarded to the **top 35%** of students at USTC.

ACTIVITIES & LEADERSHIP

Origami Club, USTC | President Jun. 2021- Jun. 2022

- Designed and organized the first origami exhibition “Vividly Comes to Life on Paper” in USTC.
- Conducted origami teaching sessions twice per month, sharing the art of origami with over 200 participants.

Hefei Chunyu Parent Support Center for Intellectually Disabled Children | Volunteer Sep. 2021- Jan. 2022

- Played basketball with autistic children and helped coaches to keep order in class once per week.

ADDITIONAL INFORMATION

Language: Chinese (Mandarin and Sichuanese dialect), English (TOEFL 103 with 23 in Speaking).
Programming & Software Skills: Python (mainly), C, MATLAB, R, Latex, Linux, HTML, CSS, SPSS, Office, etc.
Interests: Origami, click here to see my artwork.